

MUHAMMAD HAIDER ZAIDI

☎ +49 1523 1698988 ✉ mhaiderzaidi21@gmail.com in linkedin.com/in/muhammad-haider-zaidi
🌐 github.com/chefhaider 🏠 Bavaria, Germany 🎂 21.06.2000

Technical Skills

Programming: Python, C++, SQL, Javascript

Deep Learning: PyTorch, TensorFlow, Scikit-learn, Hugging Face, LLMs, Computer Vision, Time-series Analysis

MLOps: Docker, Kubernetes, AWS SageMaker, FastAPI, Azure DevOps, CI/CD, Automated Testing (pytest), Model Monitoring, Deployment, Real-time Systems, Backend Development

Data Engineering: PySpark, Airflow, Databricks, ETL Pipelines, Data Processing, SQL, Hadoop Ecosystem (Hive, Sqoop, NiFi)

Experience

German National Metrology Institute (PTB)

March 2024 – Present

Student Assistant Researcher

Berlin, DE

- Designed and evaluated deep learning models for metrological applications, enhancing **model performance, stability, and training efficiency** for critical operational inputs.
- Explored novel architectures and physics-informed hybrid loss functions to improve **model accuracy** and ensure **robustness** in production environments.
- Collaborated with Data Scientists and PhD researchers on advanced model development, contributing to a publication at FLOMEKO.
- Developed dashboards for **data visualization** and **model performance monitoring**, crucial for **MLOps** and **operational oversight**.

Folio3

March 2022 – Nov 2023

Machine Learning Engineer

California, US (remote)

- Optimized vitiligo segmentation accuracy by 94% and increased inference speed by 5x using AI algorithms; deployed **ML services** via **Docker** and **FastAPI**, ensuring **robust production operation**.
- Engineered a **real-time** tennis foul detection system with **TensorRT**, significantly reducing **compute cost** and **inference time** by 6x, demonstrating operational efficiency.
- Managed comprehensive **MLOps pipelines** on **AWS SageMaker**, addressing **model data drift** for YOLO models and maintaining **Kubernetes infrastructure** for scalable deployments in a **cloud environment**.
- Conducted educational sessions on machine learning and big data at universities, demonstrating ability to explain **technical topics** and foster knowledge sharing.
- Developed Proof-of-Concepts using YOLO, few-shot learning, and CNN for client demos, showcasing diverse **ML concepts** and rapid prototyping.

Manda GmbH

March 2024 - June 2024

Werkstudent Python Developer

Nürnberg, DE

- Executed **full-stack development** for a website offering **AI-based services** using Flask, HTML, and CSS, contributing to **backend service** implementation.
- Developed an **AI chatbot application** leveraging the OpenAI API and relevant LLM tools/frameworks, integrating **ML services** into a user-facing product.

Freelance

July 2020 – March 2022

Data Engineer

Remote

- Designed and implemented **ETL pipelines** for acquiring and processing structured and unstructured data, foundational for **data pipelines** in **ML workflows**.
- Automated **data ingestion** and cleaning workflows for client analytics use cases, enhancing data reliability and efficiency.

Systems Ltd. - Visionet

May 2021 – Feb 2022

Data Engineering Intern

Lahore, PK

- Gained practical experience in the **Hadoop ecosystem**, including Hive, Sqoop, NiFi, **PySpark**, and **Airflow**, supporting the development of **data pipelines** and **training pipelines**.
- Authored a learning blog on data engineering concepts, demonstrating ability to explain **technical topics** and share knowledge effectively.

Projects

CAFA 6: Protein Function Prediction

2022

Deep Learning Competition

- Architected a weighted-probability ensemble integrating DPFunc and DeepGOPlus, utilizing AlphaFold for advanced **feature extraction** and taxonomic bias mitigation to improve **model accuracy**.
- Achieved a 4% increase in **model accuracy** through custom **feature engineering** across MF, CC, and BP namespaces, demonstrating strong ML and **evaluation** skills.

Razanlytics: Real-time Bitcoin Price Forecasting

2022

Thesis Project

- Developed a **real-time** Bitcoin price forecasting system, integrating time-series models for predictive analytics, relevant for **operational estimators**.
- Implemented robust **training** and **inference paths** for model deployment and continuous **retraining** to ensure model relevance and **stability**.
- Leveraged cloud resources for scalable data processing and model hosting, showcasing experience with **cloud environments** for **ML services**.

Saaz: AI Self-Playing Chitralli Sitar

2023

Signal Processing, Music Generation

- Designed a **real-time** signal processing system integrating temporal dependencies for music generation, relevant for **operational estimators** and **ML services**.

- Implemented transformer-based models for sequence prediction in audio generation tasks, showcasing advanced **ML model development**.

Hugging Face

Open-Source

Contributor

- Co-authored the Zero-shot Image Classification technical documentation and implementation page on the Hugging Face Hub, demonstrating ability to explain **technical topics** and ML concepts.

Education

Friedrich Alexander Universität

Present

Master of Science in Artificial Intelligence

Erlangen, DE

- Specialized in deep learning for computer vision and spatio-temporal modeling, relevant for **tour planning** and **operational estimators**.
- Relevant coursework: Advanced Neural Networks, Computer Vision, Autonomous Systems.

National University of Computer and Emerging Sciences Aug 2018 – June 2022

Bachelor of Science in Computer Science

Karachi, PK

- Dean's List of Honor
- Specialized in Machine Learning and Computer Vision with focus on time-series analysis, providing a strong foundation for **ML concepts** and model development.

Certifications

PyTorch: Deep Neural Networks with PyTorch | **Coursera:** TensorFlow Developer Certificate
| **Databricks:** Lakehouse Fundamentals | **Datacamp:** Computer Vision in Python